Lecture 5 DSDM (Agile Project Framework)

COMP1787 Requirements Management

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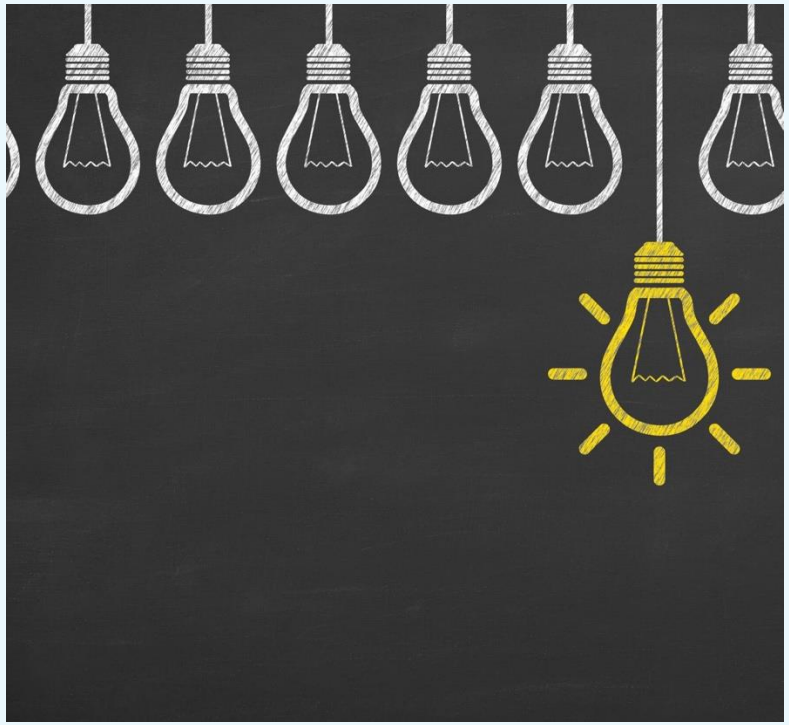
What is DSDM? (1)

- A framework for RAD development
- **Dynamic Systems Development Methodology (DSDM)** has evolved as the result of efforts by the **Dynamic Systems Development Consortium now rebranded as DSDM Agile Project Framework (2014) (https://www.agilebusiness.org)**
- Formed in January 1994 by over 300 organisations such as Barclays, Norwich Union and Logica.
- Their brief was to encourage the take up of RAD development techniques in a structured & systematic way.



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What is DSDM? (2)

- There have been several versions since the early days.
  - Evolved from DSDM Atern to the current
- The **DSDM Agile Project Framework (**2014) is the backbone for,
  - Agile Project Management (AgilePM®) certification (APMG accredited Agile Project Management) [(https://agilebusinessproducts.org/products/agilepgm.html](https://agilebusinessproducts.org/products/agilepgm.html))

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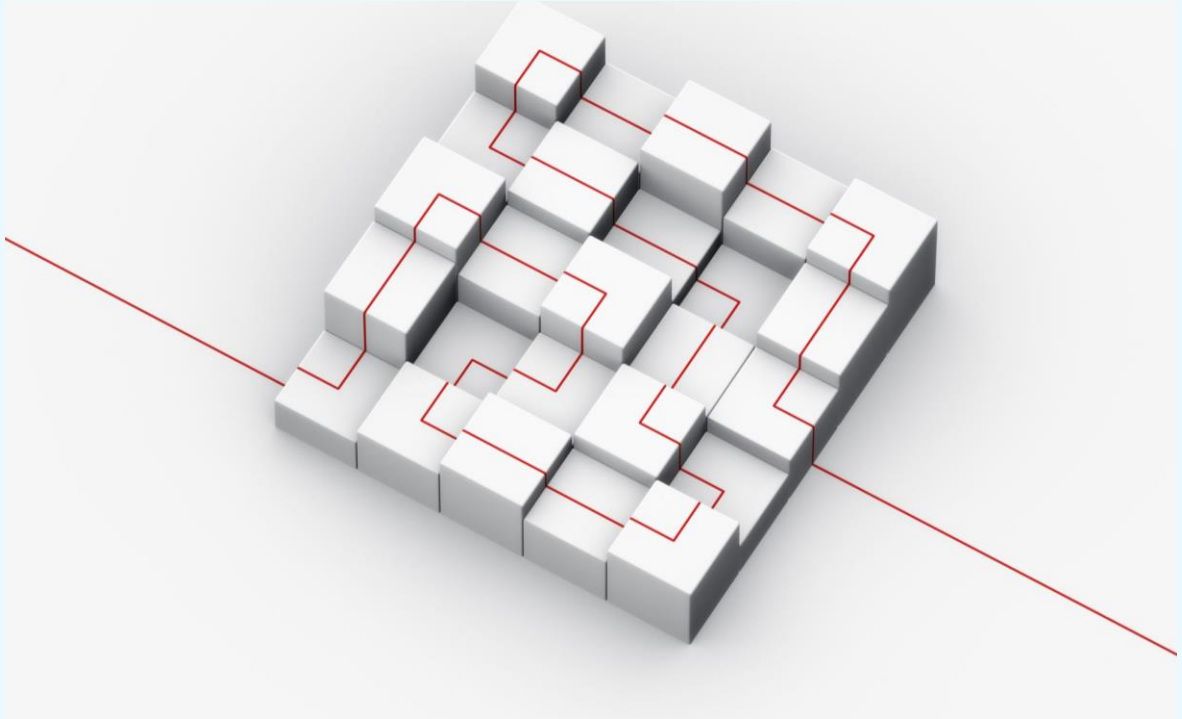
The DSDM Agile Project Framework to RAD (1)

- Provides a framework for rapid development within a **controlled** project environment.
- It does not tie the development to particular tools and techniques
  - e.g. may use data modelling, object modelling to achieve the design - DSDM is not concerned with this.



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The DSDM Agile Project Framework approach to RAD (2)

- Users must be an integral part of the development process.
- The DSDM approach aims to address the needs of all parties concerned with the development, Developers, Users, Project managers, QA Personnel, etc.
- DSDM is designed to be easily tailored and used in conjunction with traditional methods such as PRINCE2® or to complement other Agile approaches such as Scrum.

Why DSDM Agile Project Framework ?

- Developed by RAD practitioners
- It has been tested out on real, live projects

DSDM was launched to address the problems of the traditional approach (too slow, too big, not transparent enough, not enough on-going business involvement)

The problems introduced by RAD (focus mainly on speed and quick fixes, no view of the big picture issues).

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DSDM Agile Project Framework Philosophy

Best business value emerges when projects are aligned to clear business goals, deliver frequently and involve the collaboration of motivated and empowered people

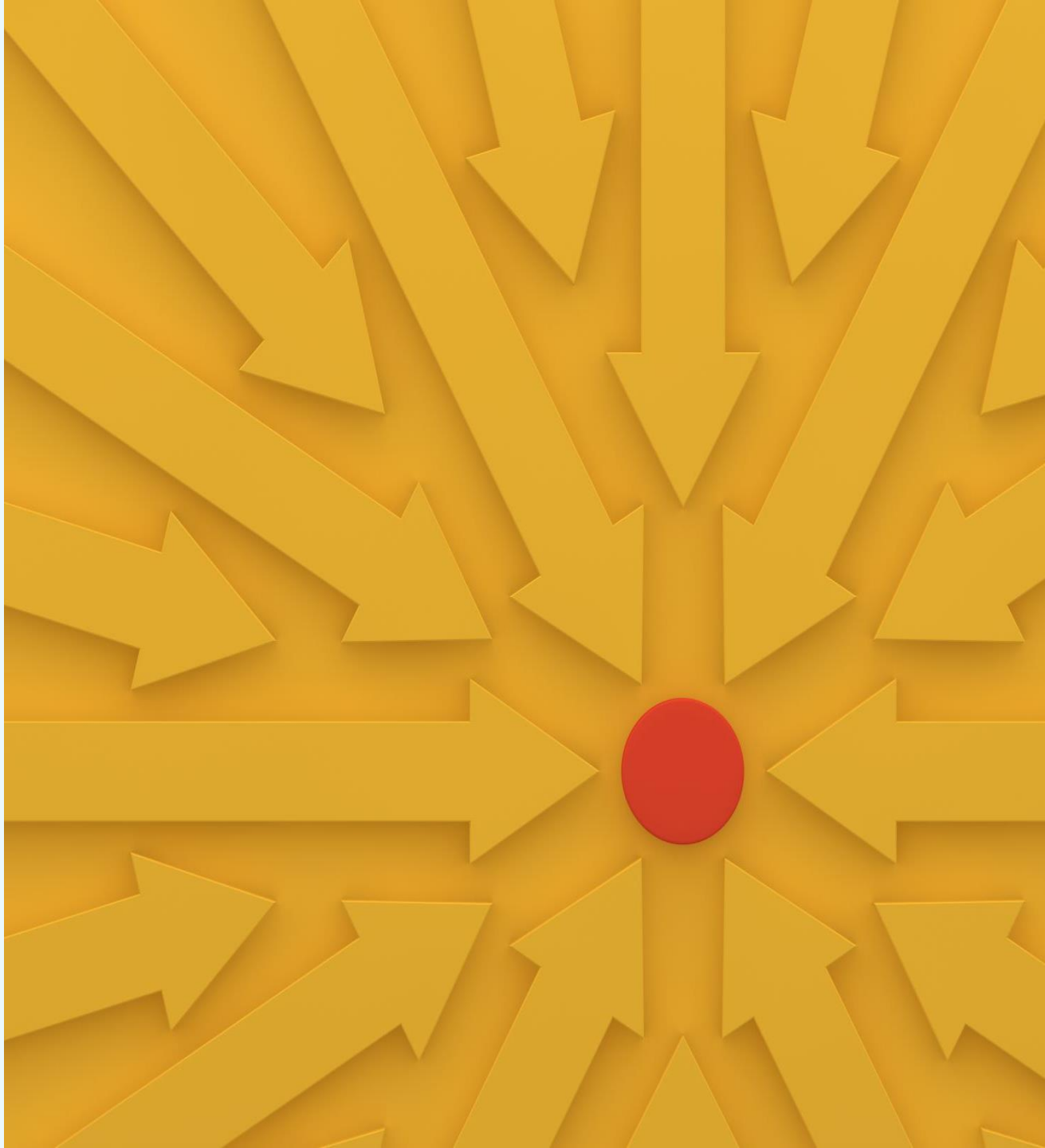
- Achieved when stakeholders:
  - Understand and buy into the business vision and objectives
  - Are empowered to make decisions within their area of expertise
  - Collaborate to deliver a fit for purpose business solution
  - Collaborate to deliver to agreed timescales in accordance with business priorities
  - Accept that change is inevitable as the understanding of the solution grows over time

The principles

- The concept of a ‘principle’ is unique to DSDM
- The principles direct the team *in the attitude they must take and the mindset they must adopt in order to deliver consistently.*
- The idea is that if you don’t follow all the principles then the success of the project will be compromised.

Principle 1:

- **Focus on the business need**
  - What is the business case?
  - What is the business goal?
    - We need to deliver what the business needs when it needs it – that's our overriding goal.



Principle 2:

- **Deliver on time**
  - Delivering products on time is quite often the single most important success factor for a project.
  - To ensure that a product is delivered on time, DSDM team will:
    - timebox approach
    - Focus on business priorities
    - Always hit deadlines
    - Build confidence through predictable delivery

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Principle 3:

- **Collaborate**
  - Team working is everything!
    - *Teams that work in a spirit of active cooperation and commitment will always outperform groups of individuals working only in loose association.*
- *To meet this principle DSDM team must;*
    - *Involve the right stakeholders, at the right time, throughout the project*
    - *Encourage pro-active involvement from the business representatives*
    - *Ensure that all members of the team are empowered to take decisions on behalf of those they represent*
    - *Build a one-team culture*
    - *Actively involve the business reps (users)*

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Principle 4:

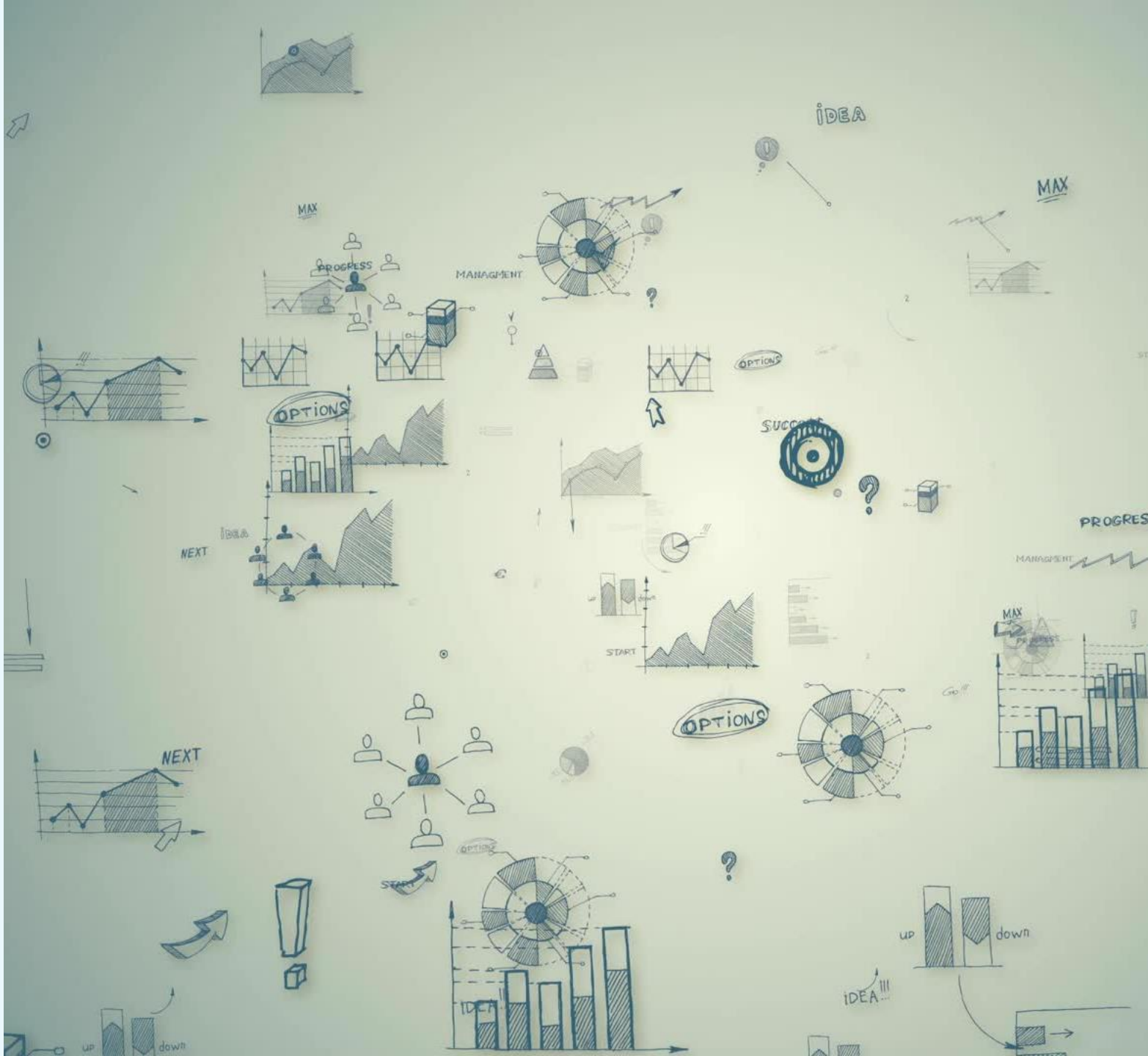
- **Never compromise quality**
  - The level of quality should be agreed at the start.
  - All work should meet that quality standard
    - Remember ‘fit for purpose’
  - To fulfil this principle the teams must:
    - Set level of quality at beginning
    - Design, document, test appropriately
    - Constantly review product development
    - Test continuously (remember prototyping)

Principle 5:

- **Build incrementally from firm foundations**
  - the team must understand the scope of the business solution and the proposed solution, *but not in such detail that the project becomes paralysed (define the high-level requirements)*
  - Carry out appropriate and enough design up-front (EDUF) to create strong foundations
  - Formally re-assess priorities and informally re-assess ongoing project viability with each delivered increment

Principle 6:

- **Develop iteratively**
  - Only way to *converge on an accurate business solution* is to refine by backtracking (iterative approach)
    - *It is very rare that anything is built perfectly first time*
    - *Projects operate within a changing world.*



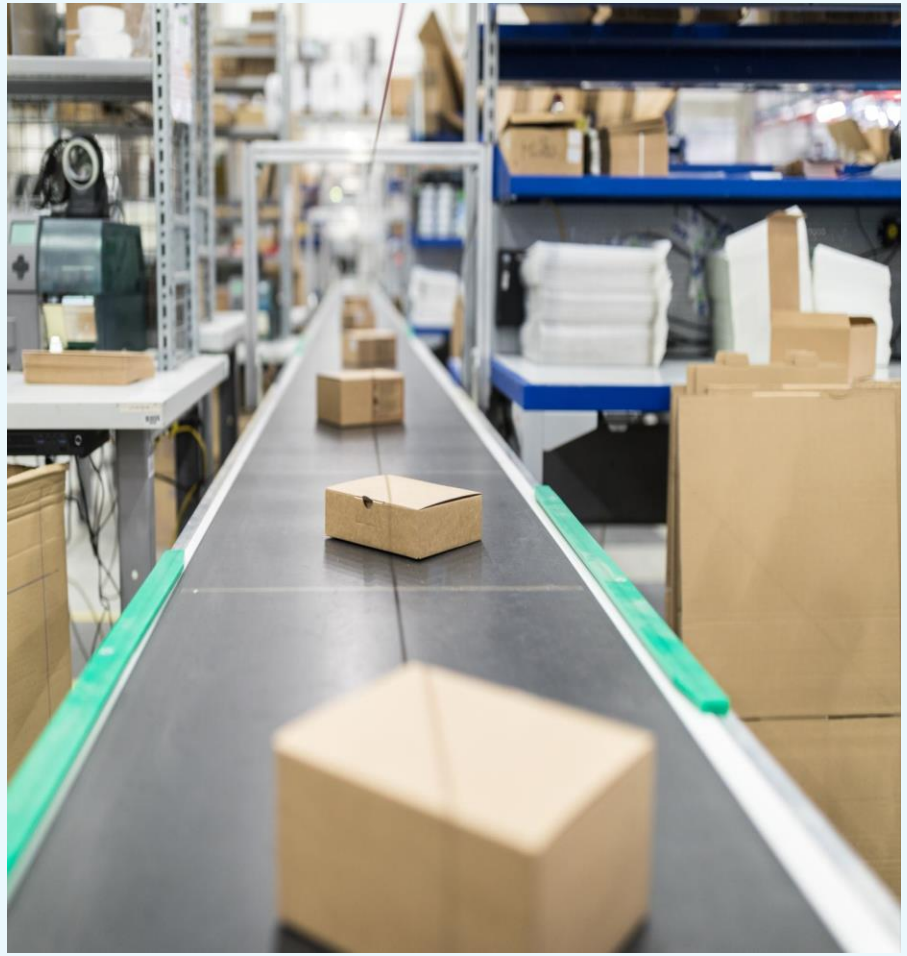
Principle 7:

Communicate continuously and clearly

- Poor communication is often cited as the biggest cause of project failure.
- Teams will:
  - Run daily stand-up sessions
  - Use workshops with a facilitator
  - Use visual communication techniques such as modelling and prototyping
  - Keep documentation lean and timely
  - Encourage informal, face-to-face communication

Principle 8:

- **Demonstrate control**
  - The project must be always controlled
    - Iterative, incremental approach using prototyping requires close monitoring to ensure success
    - To maintain confidence control must be always demonstrated:
      - Plans and progress visible to all
      - Measure progress through focus on delivery of products rather than completed activities (product-based approach rather than activity-based approach)
      - Manage proactively
      - Evaluate project viablity based on business objectives



Pre-Project

Only right projects are started, and that they are set up correctly, based on a clearly defined business objective.

Feasibility

Just enough technical and business perspective viability

Foundations

Understand the scope of work, how it will be carried out, by whom, when and where.

Deploying the first release of many

Evolutionary Development

Assemble

Review

Deploy

Deployment

Post-Project

Iterative Development, timeboxing, and MoSCoW prioritisation with Modelling and Facilitated Workshops

The DSDM Process

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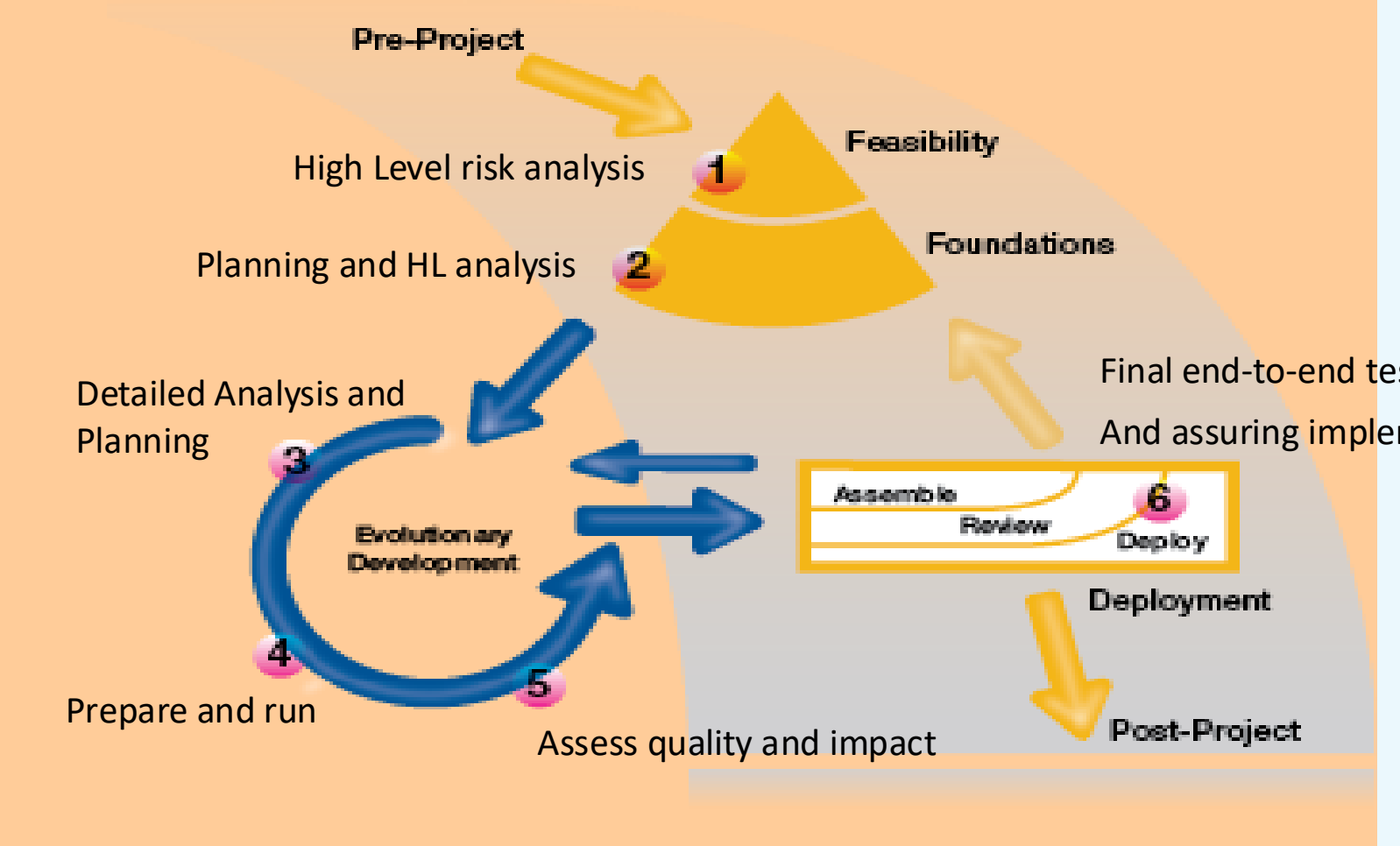
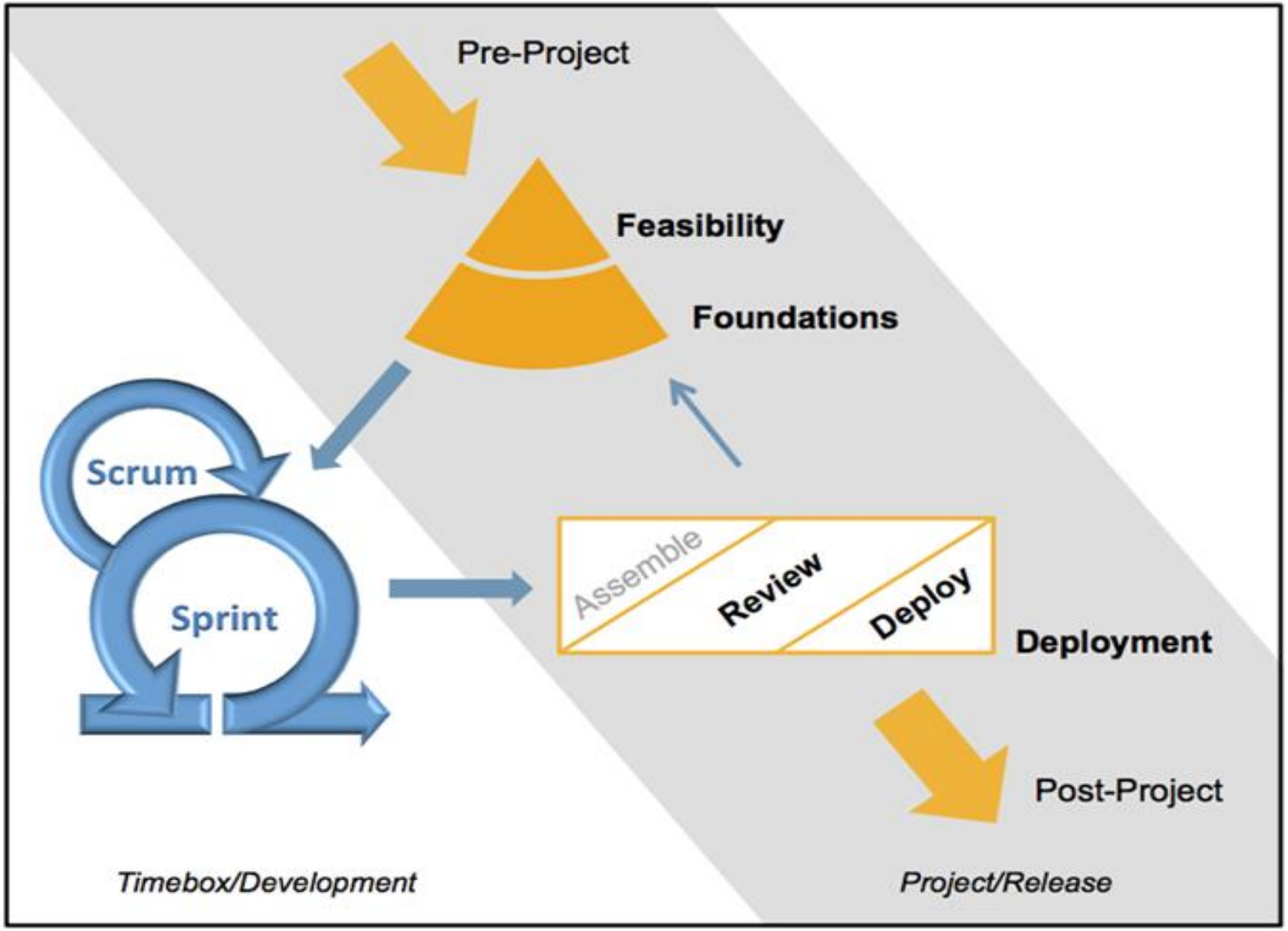
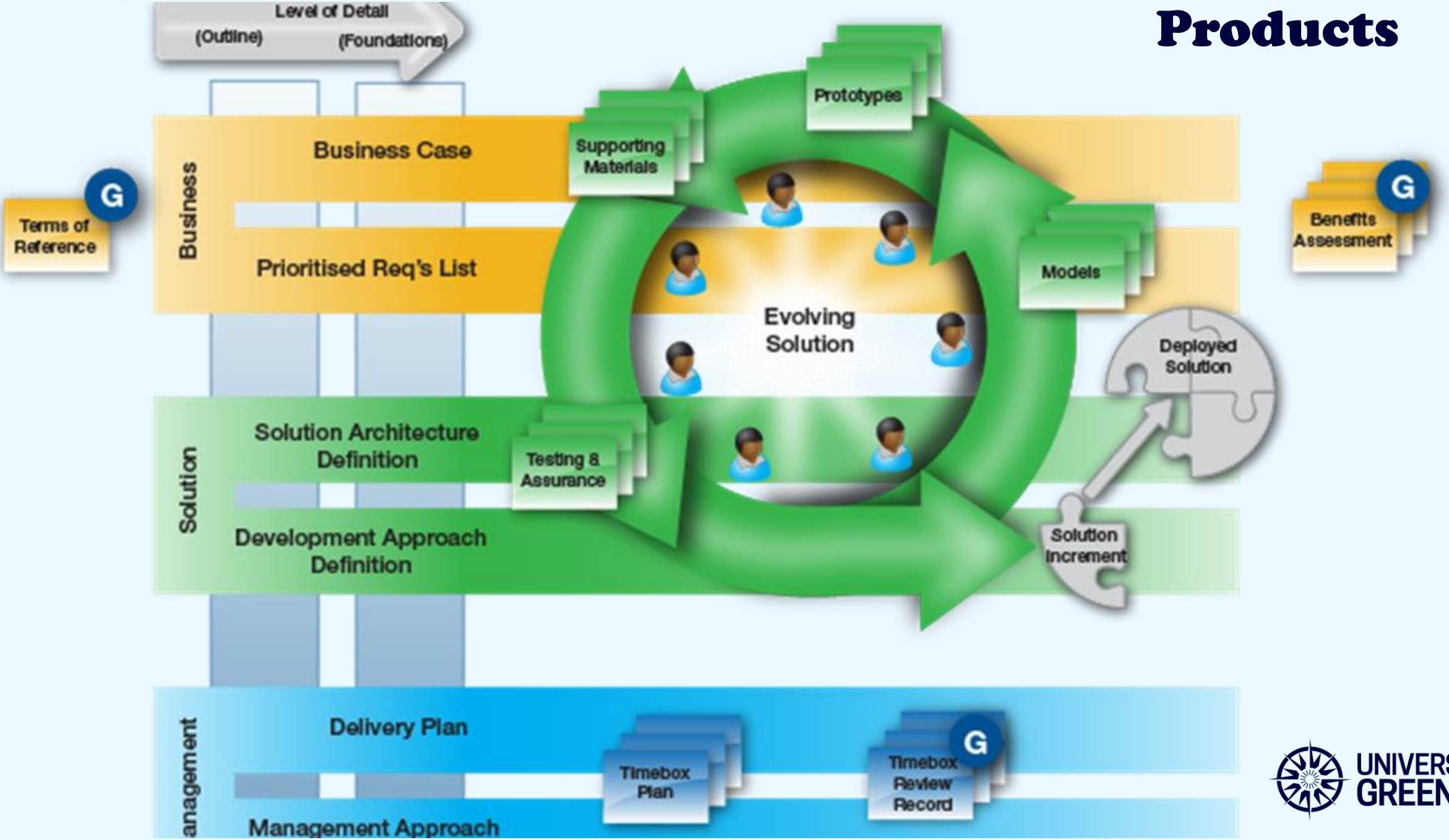
Planning and Quality



DSDM Agile Project Framework with Scrum



Products



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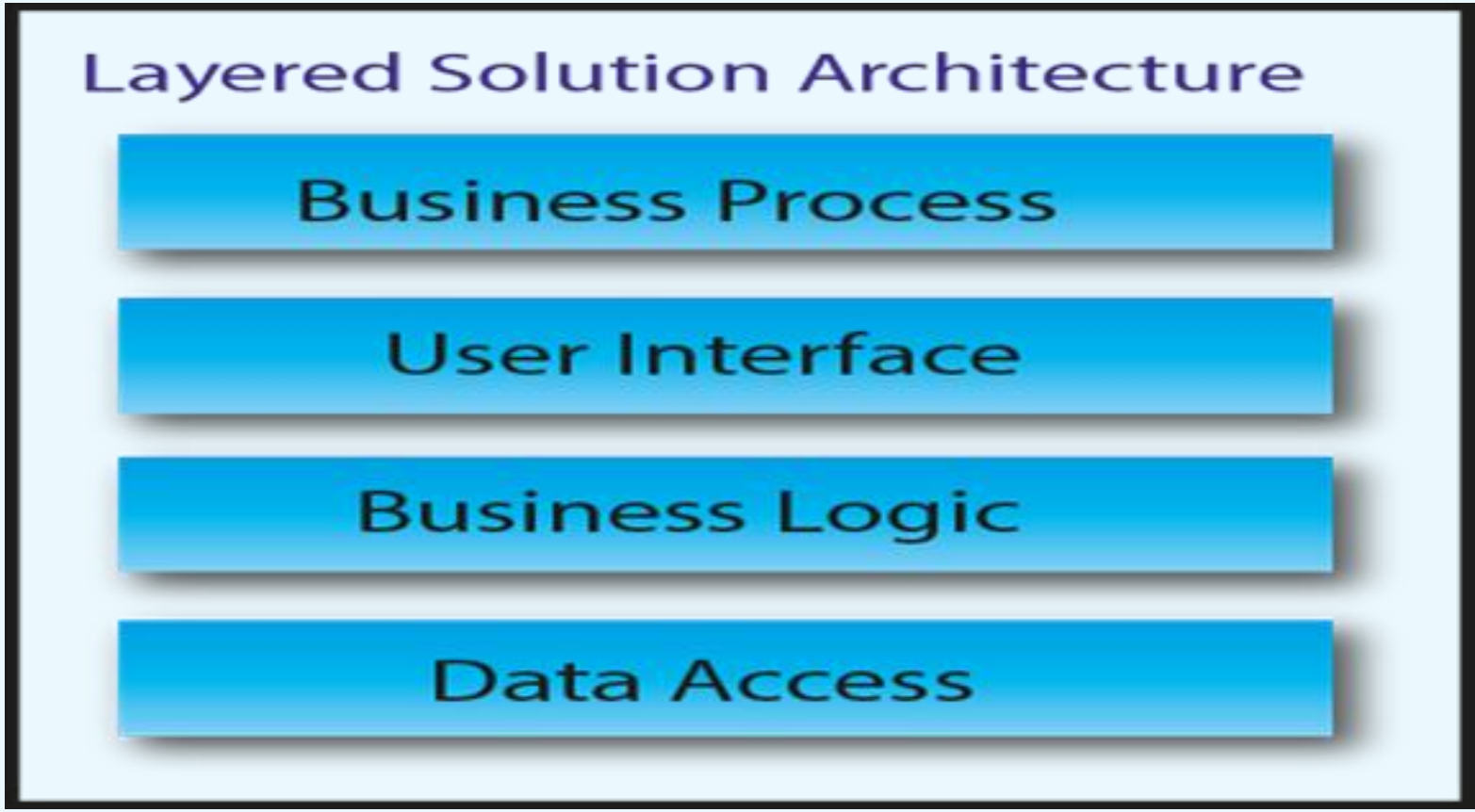
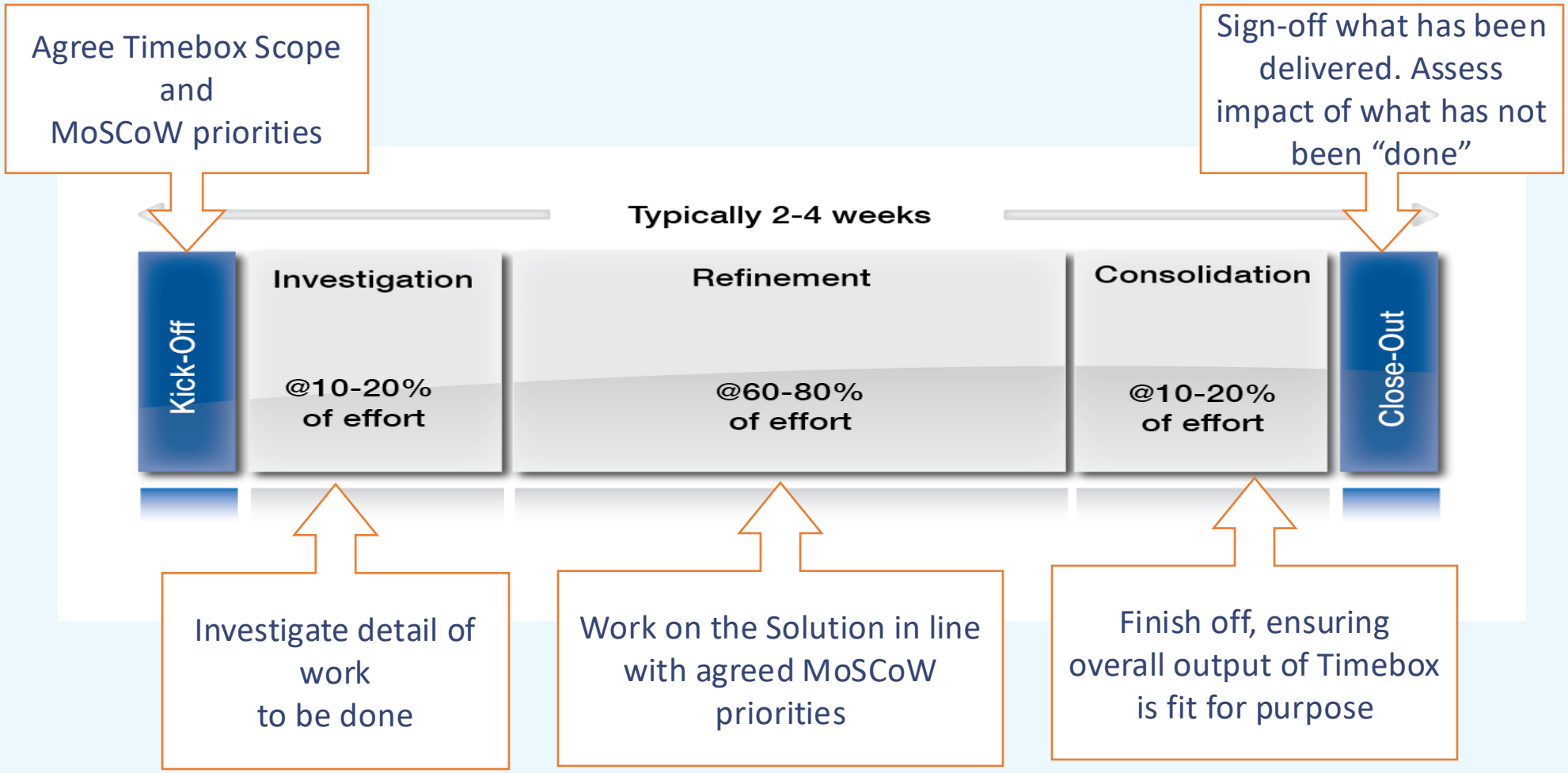
Iterative Development

- Solution Focus

- At any given time, the Iterative Development cycle may be applied to an evolution of the solution from a:
  - Functional perspective:
  - Usability perspective:
  - Non-functional perspective:

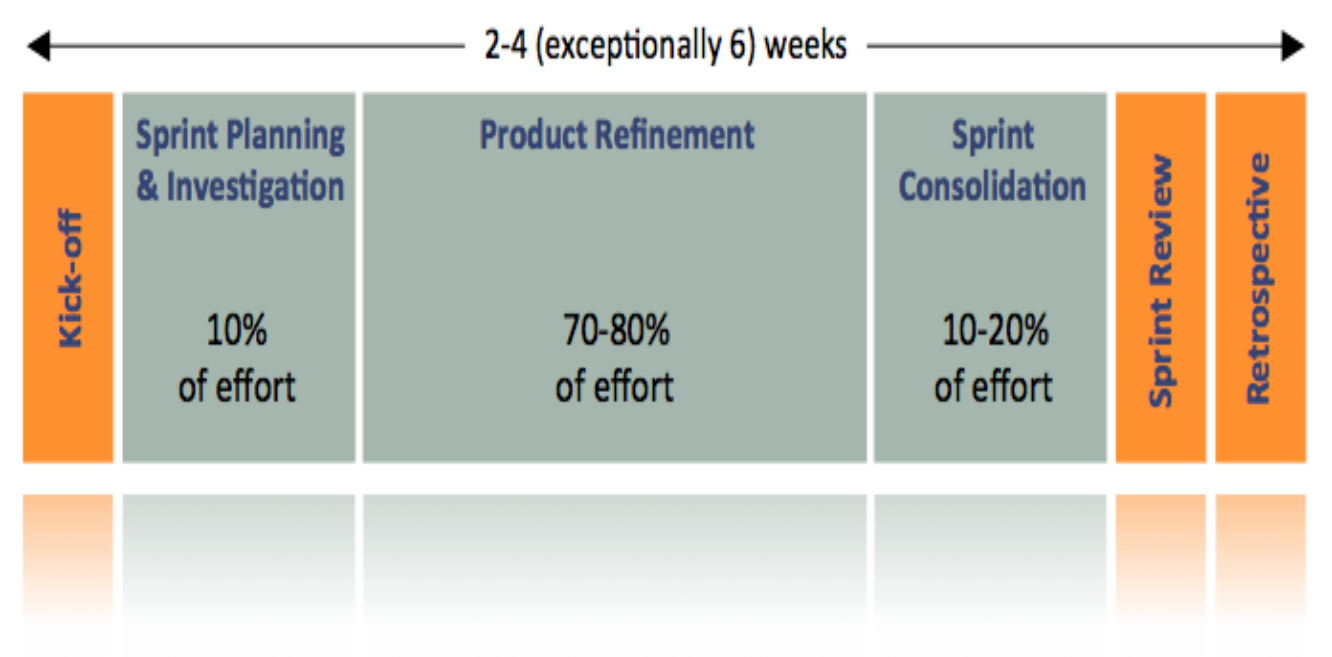


Structured Timebox



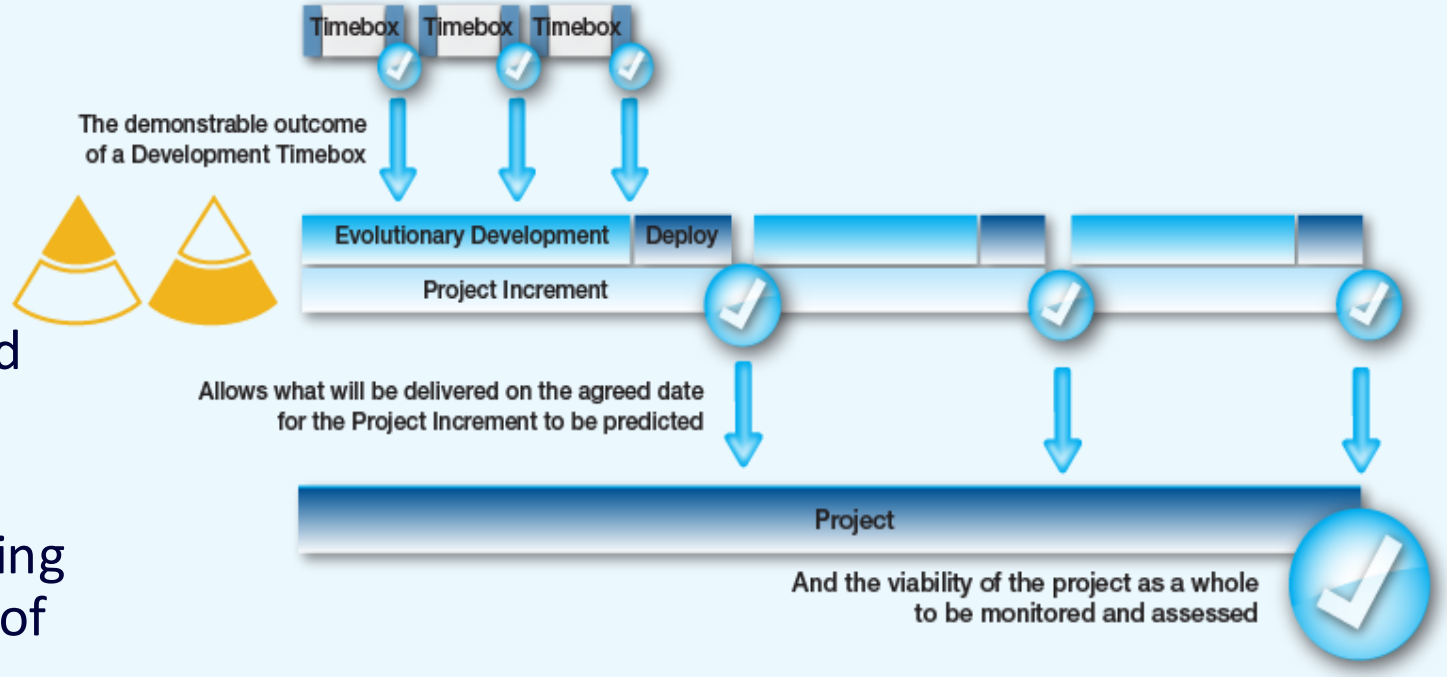
Free Format Timebox

Timeboxing and Scrum

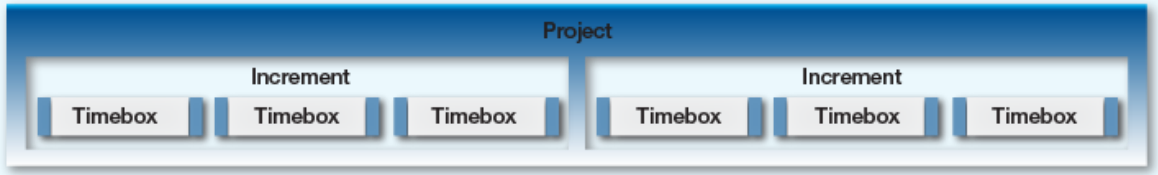


Planning Concepts

- Outcome-based planning and empowerment
  - PM responsible for high-level collaborative planning for incremental delivery of the business solution
  - SDT responsible for detailed work in timeboxes
- Planning horizons
  - Long-term outline (Delivery Plan)
  - Short term detail (Timebox Plan)



Timeboxing



- Factors to consider when scheduling a Timebox:
  - Business priority
  - Business and technical risk
  - Solution architecture and external dependencies
  - Ease of implementation and the drive for an early return on investment
  - Availability of critical or specialist resources
  - Constraints associated with business process or corporate policy
  - Dealing with change

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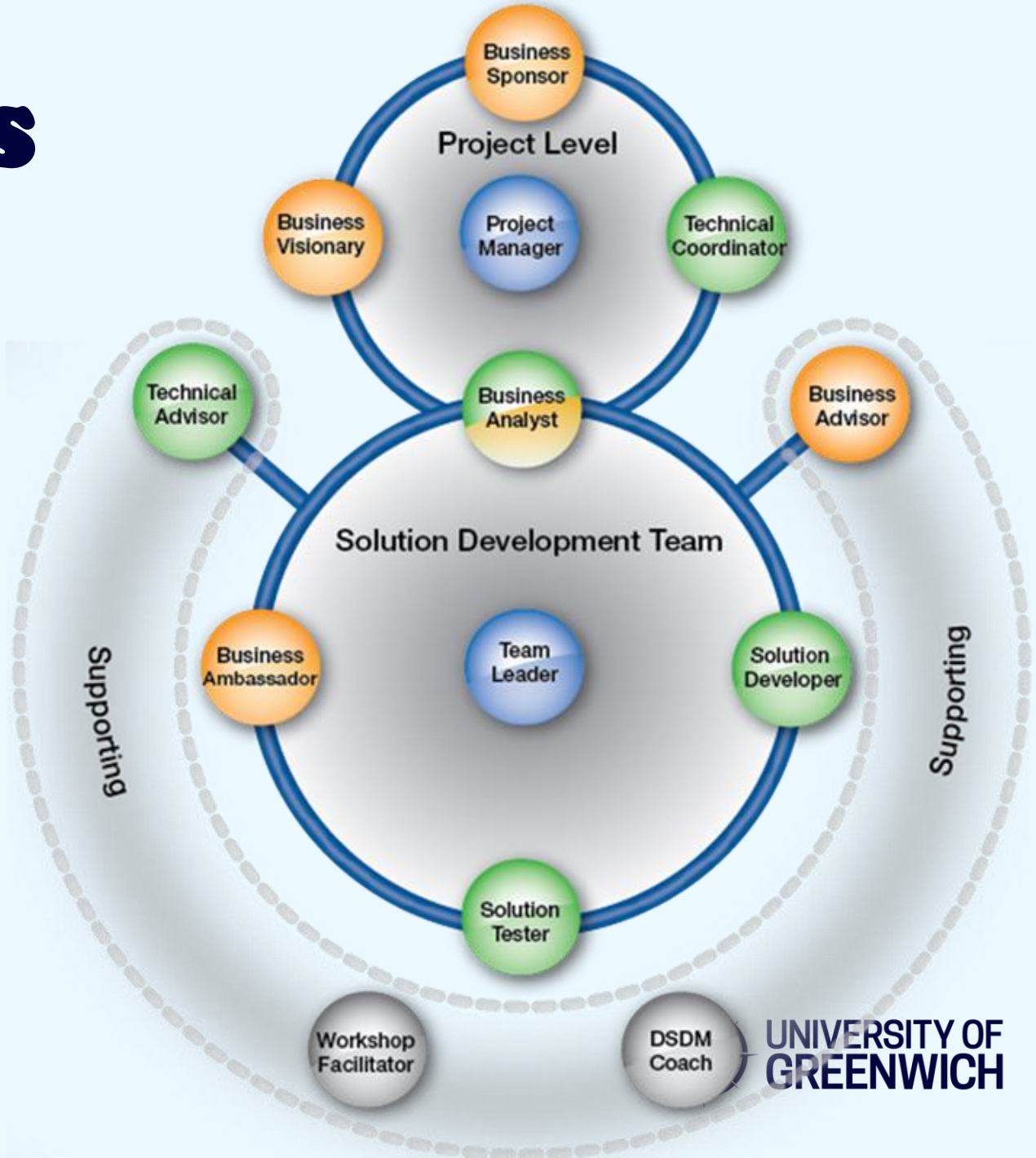
Teams and Roles

- Orange – Business Interests
- Blue – Project & Management Interests
- Green – Solution / Technical Interests
- Grey – Process Interests
- One role does not necessarily mean one person
- One role may be split between two or more people



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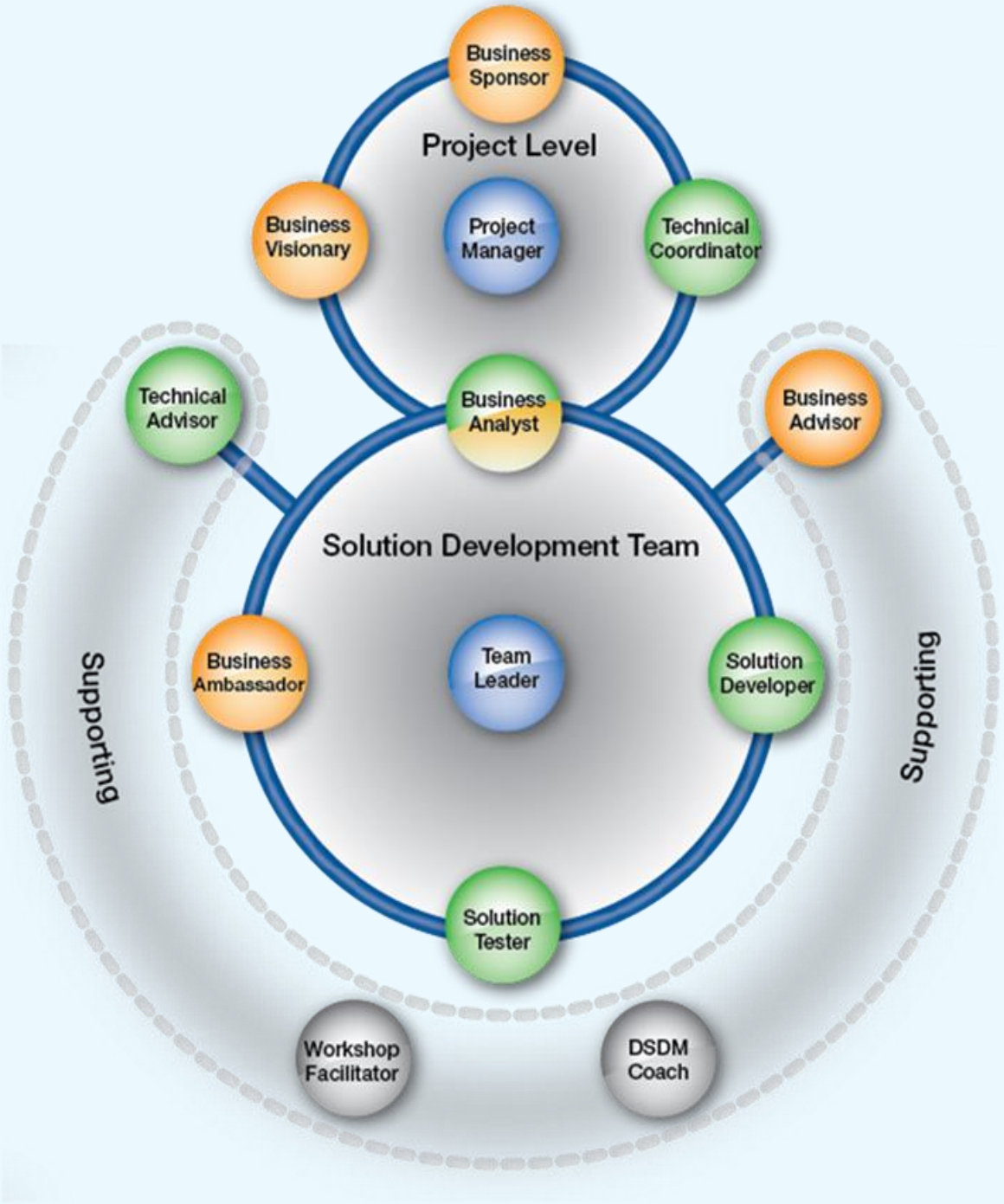
Business Sponsor

- Owns the project's business case
- Maintains its viability
- Managing the budget and resources
- Facilitating swift decision-making for escalated issues, resolving conflicts promptly, empowering team members, and staying updated on progress and challenges.

Business Visionary

- Defining the business vision for the project
- Communicating and promoting the business vision to all stakeholders
- Monitoring progress of the project in line with the business vision
- Owning the wider implications of any business change from an organisational perspective
- Contributing to key requirements, design and review sessions
- Identifying and owning business-based risk
- Defining, and approving changes to, the high-level requirements
- Ensuring collaboration across stakeholder business areas
- Ensuring business resources are available to the project
- Promoting the translation of the business vision into working practice
- Empowering the business roles within the Solution Development Team
- Acting as an arbiter of business differences



Business Ambassador

- Contributing to all requirements, design and review sessions
- Providing the business perspective for all day-to-day solution development decisions
- Providing the detail of business scenarios to help define and test the solution
- Communicating with other users
- Providing day-to-day assurance that the solution is evolving correctly
- Organising and controlling business acceptance testing of the solution
- Taking responsibility for the creation of the business user and support documentation
- Ensuring business participants in the Deployed Solution are properly trained and supported

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- Providing specialist input into relevant:
  - Requirements, design and review activities
  - Day-to-day project decisions
  - Business scenarios to help define and test the solution
- Providing specialist advice on, or help with:
  - Developing business user and support documentation for the ultimate solution
  - Deployment of the solution releases into the business, as appropriate

Business Advisor

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Business Analyst Role

- This role supports the Business Visionary by modelling the organisation's present and future, formulating solution options, and assisting with Business Case and Benefits Assessments.
- It ensures clear communication, quality requirements, and manages related documentation effectively, including PRL interpretations.

Business Analyst Role cont.

- Ensures the thorough modelling and consideration of business implications during solution evolution
- reviews the impact of decisions and ensures cohesion between business and technical components.
- It also oversees non-functional requirements and tracks business requirements to acceptance.
- Additionally, it collaborates with the Business Visionary to support solution implementation into live use.

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Project Manager

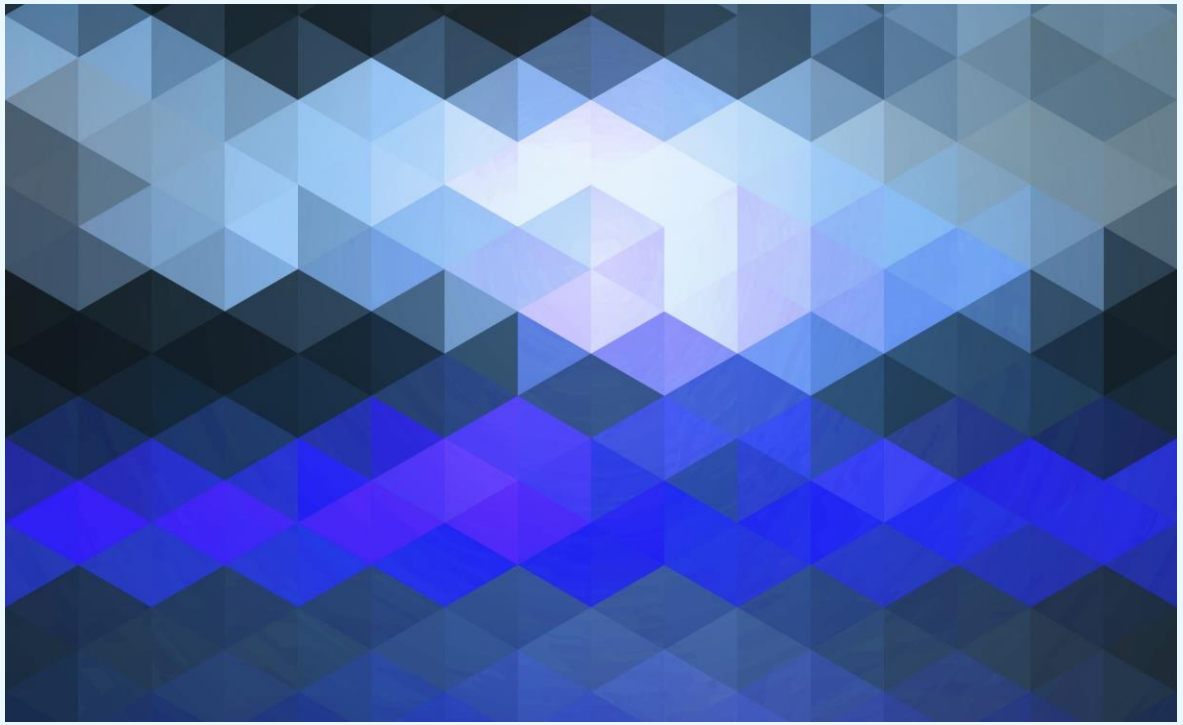
- This role focuses on effective communication with stakeholders, high-level project planning, and collaborating on the Delivery Plan with the SDT.
- It monitors progress, manages risks and issues, and ensures team empowerment and motivation.
- Additionally, it handles escalated problems, provides guidance to the SDT, and attends Daily Stand-up meetings as needed.

Technical Coordinator

- This role focuses on controlling technical architecture and environments, coordinating technical activities, and managing technical risks.
- It advises on non-functional requirements, estimations, and promotes technical best practices.
- Additionally, it approves the solution's technical fitness before deployment, manages the transition to live use, empowers technical team members, and resolves technical differences.

Technical Advisor

- Requirements, design and review sessions
- The operational perspective for day-to-day decisions
- Operational or support scenarios to help define and test the solution
- Assurance that the solution is evolving correctly
- Operational acceptance testing
- Development of technical support documentation
- Training of technical operations and support staff
- Incremental Deployment of the solution releases, as appropriate



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Team Leader

This role facilitates the team's focus on delivering products on time, encouraging full participation, and ensuring proper control of the Iterative development process.

It schedules and oversees testing and review activities, manages risks and issues at the Timebox level, and monitors daily progress.

Additionally, it facilitates communication with the Project Manager, conducts timely and focused Daily Stand-ups, and organizes reviews and retrospectives with the team.

Solution Developer

This role collaborates with all Solution Development Team members to iteratively develop the Solution Increment and necessary models.

It tests its own work before independent testing, adheres to technical constraints and organizational standards, and participates in quality assurance efforts.

Additionally, it records and interprets changes to detailed requirements, interpretation, and information affecting the solution's evolution.

Solution Tester

- Carrying out all types of technical testing of the solution as a whole
- Liaising with the Business Analyst and Business Ambassador to help clarify acceptance criteria for requirements
- Creating test products as appropriate, e.g. test cases, test plans and test logs
- Reporting the results of testing activities to the Technical Coordinator for quality assurance purposes
- Keeping the Team Leader informed of the results of testing activities
- Assisting the Business Ambassador(s) and Business Advisor(s) in their testing

DSDM Coach

- Providing detailed knowledge and experience of DSDM
- Tailoring the DSDM process to suit the individual needs of the project and the environment in which the project is operating
- Helping the team use DSDM practices and helping those outside the team appreciate the DSDM philosophy and values
- Helping the teamwork in the collaborative and cooperative way typical of DSDM and all Agile approaches
- Building DSDM capability within the teams at all levels

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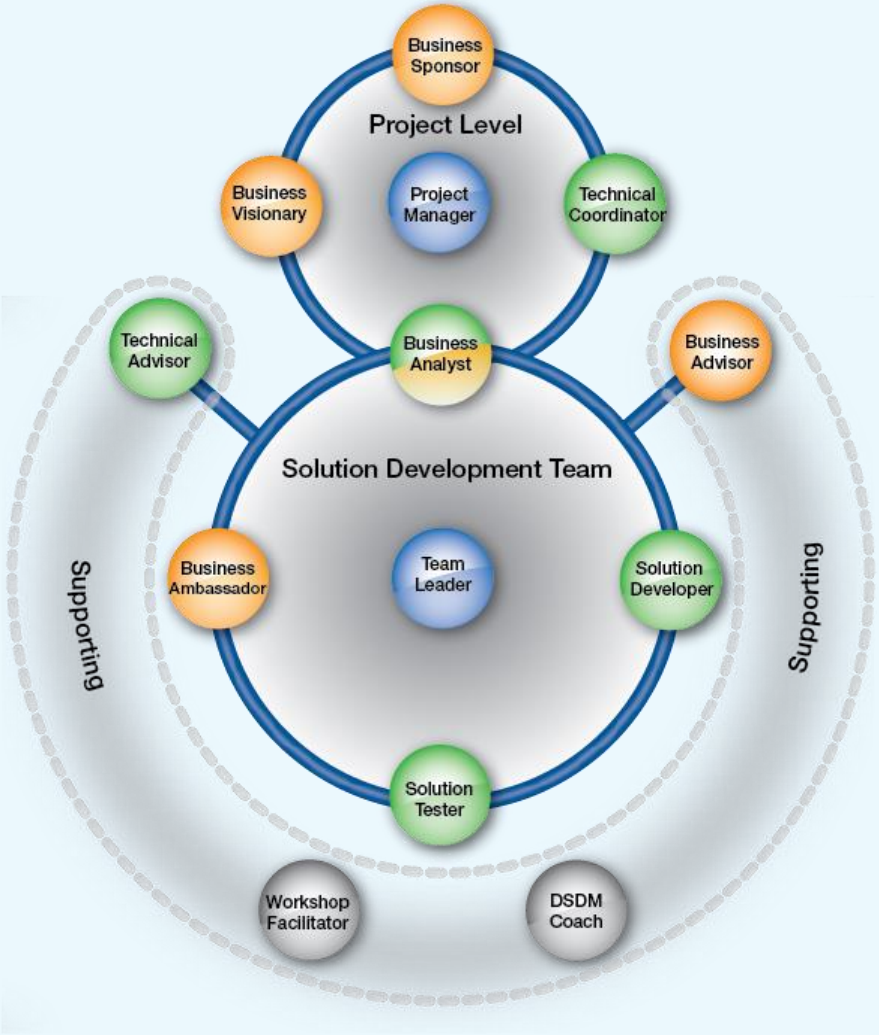
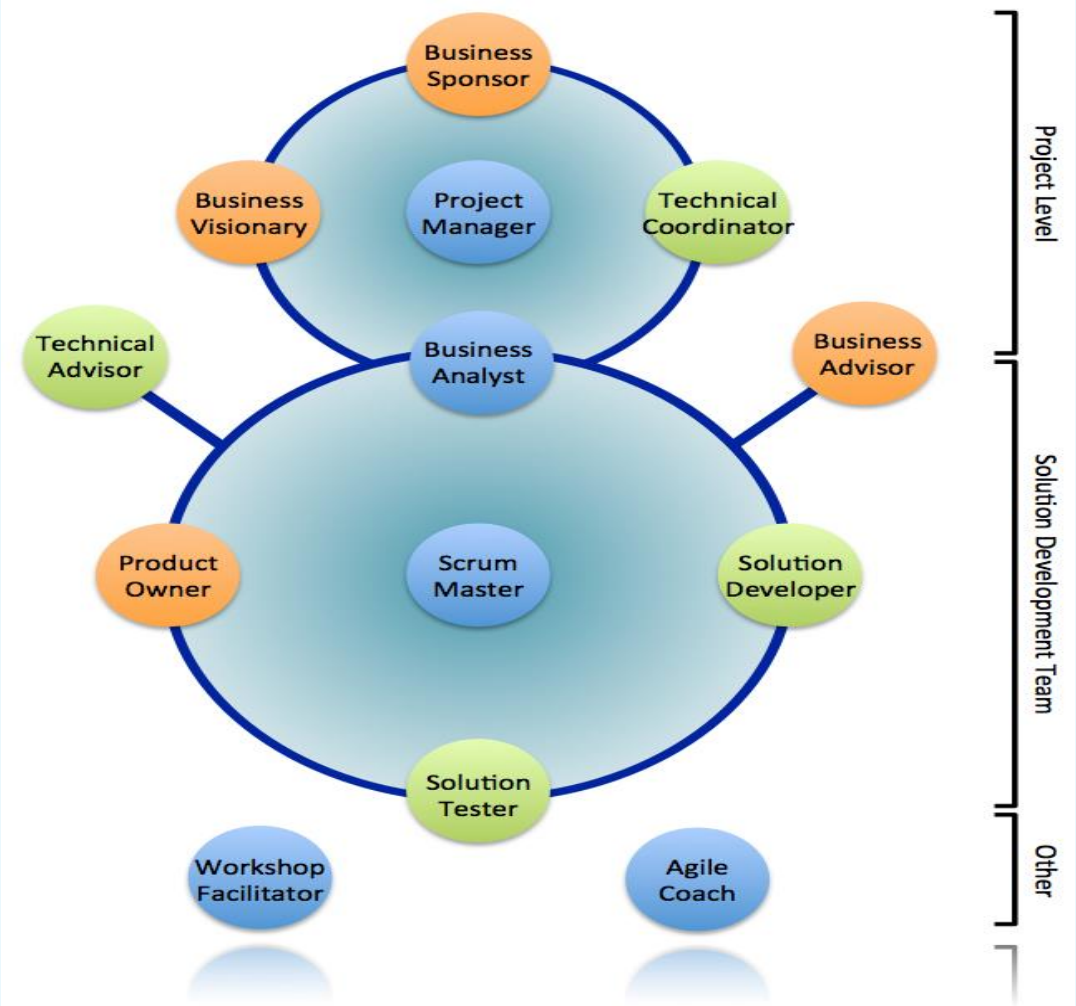
Workshop Facilitator

- This role involves preparing for workshops by agreeing on scope, planning the agenda and decision-making process, and familiarizing oneself with the subject.
- It engages with participants beforehand to confirm suitability, ensure understanding of objectives, address concerns, and encourage preparation.

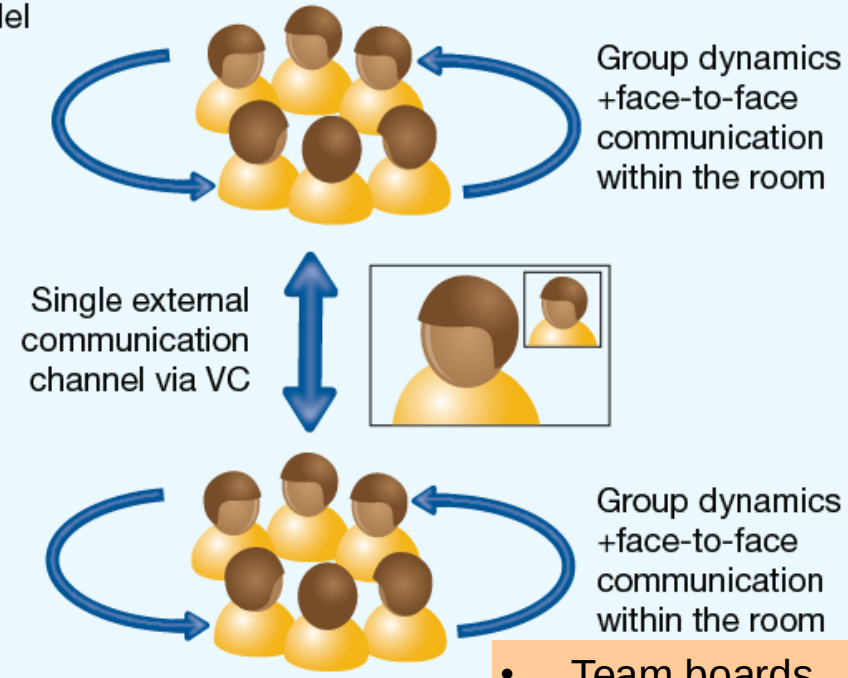
Agile PM Team Roles incorporating Scrum Roles





Planning Communication

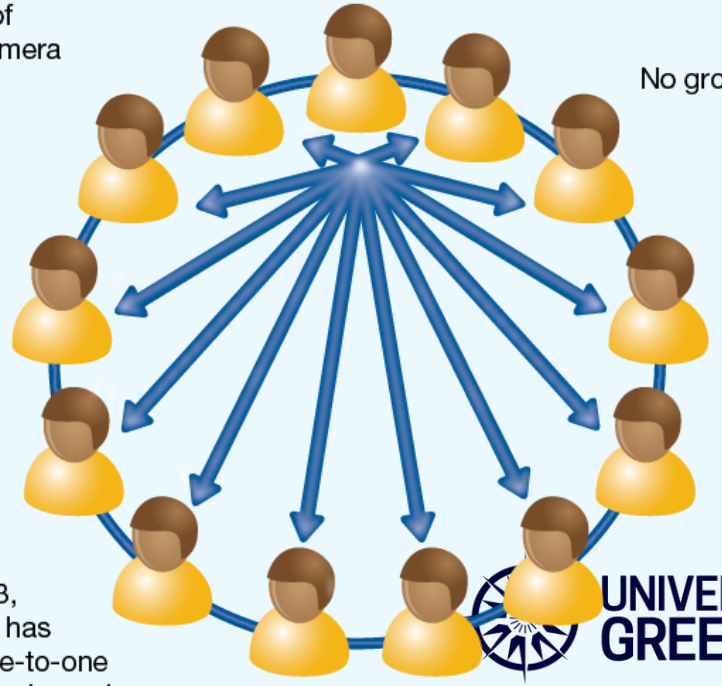
Recommended distributed team communication model



- Face-face
- (Video) Conferencing
- On-line chat
- Email
- Collaborative workspaces
- Documents

- Team boards
- Daily Stand ups
- Co-located teams
- Distributed Teams
- Collaborative workspaces
- Documents

Disadvantages of individual PC camera conferencing



No group dynamics

In a group of 13, each individual has 12 separate one-to-one communication channels

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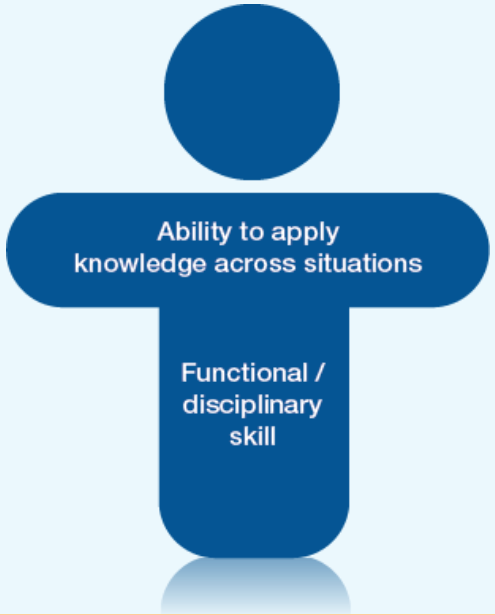
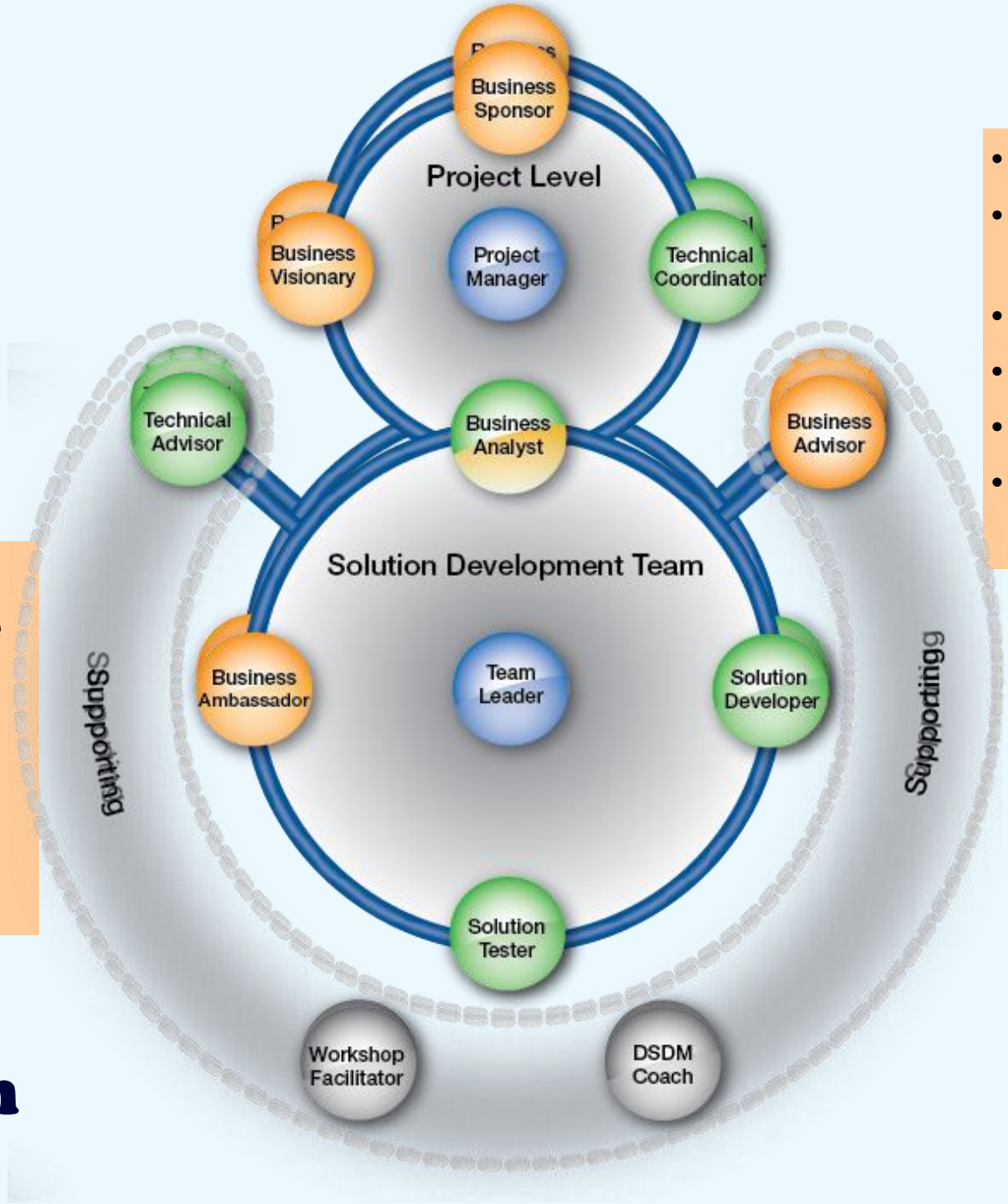


- Trust, respect
- Open minded, approachable
- Available, open to change
- Clear direction, stable
- Belief in self and others
- Good leadership, good communication

DSDM Collaboration



- Listening
- Cooperation, compromise, negotiation
- Open, honest, transparent
- Non-verbal
- Self-awareness
- Appreciation of others, empathy

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To summarise

- *DSDM approach is ideal in today's business environment*
  - *Consider techniques that can be used to support this environment*
- DSDM is a method that has been developed to support the ideas behind RAD
  - Philosophy, principles, process with products, people, practices

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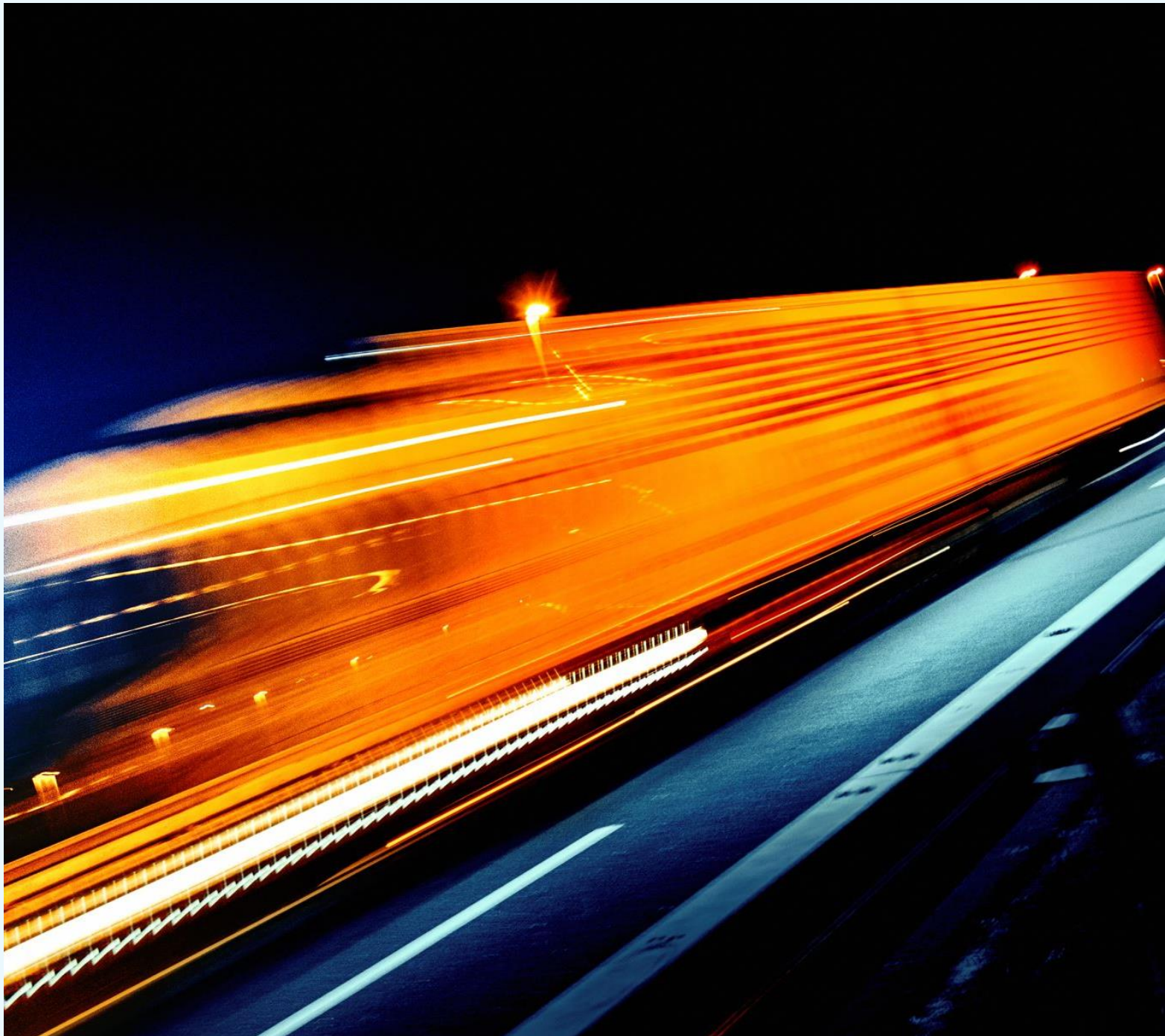
Acknowledgement

- Agile Business Consortium (https://www.agilebusiness.org)